Light Bubbles – The mechanical solution to wave-particle duality

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Abstract

Within the framework of the Universal Quantum Foam Hypothesis (UQSH), radiation is not interpreted as the transport of discrete particles through empty space, but as a fieldmechanical excitation of a continuous, elastic medium. Light corresponds to the freely propagating boundary dynamics of this medium and expands isotropically as a spherical stress front. Emission is understood as the local relaxation of previously bound field states, where energy is not carried by objects but transferred through successive reorganization of neighboring regions. To provide an intuitive description of this process, the concept of "Light Bubbles" is introduced. A Light Bubble denotes a spherically expanding field response whose propagation, interference, and detection are manifestations of continuous reorganization within the same medium. Phenomena such as interference, double-slit behavior, and apparent particle localization arise not from a wave-particle dualism, but from the combination of extended field oscillations and localized coupling to material structures. Furthermore, it is shown how bound field nodes may emerge from freely propagating excitations through geometric self-organization. Matter thus appears as stabilized, locally circulating field motion, while radiation represents the limiting case of open propagation. Black holes are interpreted as highly saturated stress regimes in which bound states lose stability and convert into free relaxation modes. This approach provides a unified ontological description of radiation, matter, and gravitational boundary states as different organizational forms of a single continuous field and offers a consistent alternative to particle-centered interpretations.

1 The End of the Emission Dogma

Radiation does not arise because an object "emits" something from itself, but because a locally bound state of tension dissolves. Whenever matter releases energy, for example in a star, in an accretion disk, or in an excited atom, a previously stored field deformation becomes unstable. The medium can no longer maintain this state in a bound form and transitions into relaxation. This relaxation itself is the radiation. Nothing is thrown outward. Instead, the field reorganizes locally and passes on the stored tension as a moving disturbance.

This process does not occur in a directed manner, but isotropically. In every elementary time step, the medium responds simultaneously in all directions. Emission is therefore spherical in its origin. What emerges is a spherical front of tension that propagates outward at the boundary rate c of the field. Each minimal pulse corresponds to a renewed update of the medium. The front shifts step by step outward, always by exactly the distance that the field can reorganize during that time. The speed of light is nothing other than this maximum rate of reorganization.

While the disturbance spreads, no substance moves through space. Instead, each region of the medium transfers its deformation to the neighboring one. Energy is handed on, not transported like a body. One may imagine this as a continuous relay. Each zone briefly takes over the oscillation and immediately passes it on. What we perceive as light is this chain of local transfers.

When several such field motions meet, they do not add like solid objects, but superpose. Tensions reinforce or weaken each other. Where two oscillations meet in phase, a larger local field deformation arises. Where they are out of phase, they partially or completely cancel each other. Interference is therefore not a mysterious contradiction, but a direct consequence of the fact that there is only a single continuous medium. The field can assume only one overall state at any given location. Multiple excitations merge there into a common response.

Over large distances, the radiation continues to change. Since the same excitation spreads over an ever growing spherical volume, the local tension density decreases continuously. The energy is distributed over more and more portions of space. The field oscillation thereby becomes weaker and longer in wavelength. This process corresponds to a gradual rarefaction of the excitation. The wave remains structurally intact, but loses local intensity and coherence. Small disturbances, scattering processes, and renewed reorganizations within the medium further smooth the structure. Over cosmological distances, radiation therefore appears stretched, weakened, and effectively longer in wavelength.

When such a diluted field motion finally encounters matter, the process begins anew. The incoming tension locally reorganizes the medium. Emission, propagation, superposition, and absorption are therefore not separate phenomena, but successive stages of one and the same field-mechanical dynamics. In this sense, radiation is not a journey of things through space. It is the continuous selfreorganization of the medium. The field responds to every disturbance with motion, carries this motion onward, superposes it with others, and finally binds it back again. Light is therefore nothing other than the rhythm of this universal transfer.

2 From Light Bubbles to Matter

Transition from free to bound field organization

If radiation within the framework of the UQSH is understood as a freely propagating oscillation of the medium, the question immediately arises how stable material structures can emerge from such an open dynamics. Put differently, the problem is how a locally bound and persistent state can arise from an extended, rarefying field motion. The conditions for the emergence of matter have been explained multiple times in previous works, monographs, and preprints. Here, the concern is only the general description up to the point where those conditions are reached.

In the classical particle picture, waves and matter appear as categorically separate phenomena. Waves are transient and spread out, while particles are regarded as compact, permanent objects. In the field-mechanical framework of the UQSH, this separation disappears. Both states correspond to different organizational forms of the same continuous medium. The difference lies not in substance, but in the degree of binding of the underlying field motion.

A free radiation excitation distributes its energy over an increasingly larger volume as the radius grows. The local tension density decreases, and the motion persists only as long as the medium allows unhindered propagation. However, when such an oscillation encounters regions of elevated field tension, strong curvature, or already existing coherent modes, the dynamics change fundamentally. The propagation is not interrupted, but reorganized.

Through superposition, interference, and repeated coupling to local field structures, propagating modes can no longer transport energy exclusively outward. Instead, regions arise in which the motion becomes increasingly localized. Field components circulate rather than continue to propagate. Open propagation gradually transitions into closed orbital motion. Energy is no longer spatially distributed, but bound within a coherent structure.

Such a state does not represent a solid body, but a stabilized resonance pattern of the medium. The field motion remains dynamic, yet spatially confined. This locally bound, selfconsistent organization corresponds within the framework of the UQSH to what is conventionally referred to as a particle.

Elementary matter can therefore be understood as the limiting case of strongly bound field motion, while radiation represents the limiting case of free field motion. Between the two there exists no discrete transition, but a continuous spectrum of different degrees of binding. Photons, plasmas, electrons, or more compact structures represent different levels of stability of the same field-mechanical dynamics.

The so-called wave–particle dualism thus appears in this view not as a fundamental division of nature, but as a perspectival description of two regimes of organization of a unified medium. Radiation corresponds to the open, propagating form of motion, matter to the locally bound and coherent form.

3 Field Nodes, Radiation, and Maximum Saturation

When one looks at radiation from the familiar particle perspective, the transition to the UQSH picture almost inevitably raises a concrete question that cannot simply be skipped. What happens to a wave when it encounters a black hole? Is it absorbed like an object that disappears into a funnel? Is it split so that one part falls in and one part continues on? Or is a wave indivisible, so that such statements make no sense at all? It is precisely at this point that one sees how deeply the old conception still operates. For in everyday language we think in things. Something arrives. It bounces off. A remainder stays behind. And if one part goes in, the other must remain outside. This logic is plausible in the particle picture. Within the UQSH framework, however, it is already a false grammar, because radiation is not a thing in space, but the mode of motion of the medium itself. Properly formulated, the question should be: How much of the field motion couples locally, and how much continues as propagation?

To truly meet the classical observer halfway, it is worthwhile to set the difference very cleanly. In conventional intuition, space is a stage. On this stage either particles or waves move. A wave is then a kind of extension of something that spreads across the stage, and a particle is a small packet that has a trajectory. The black hole in this picture is an object on the stage that attracts and swallows things. In the UQSH picture, however, space is not a stage but the carrier medium. The wave is not something that travels through space, but space itself in a state of oscillation. And the black hole is not a thing in space, but a boundary state of the same medium, that is, a region in which the field can no longer deepen locally at will and transitions into a saturated organizational mode. Once this shift is accepted, the question changes as well. One no longer asks whether an object is split, but how a moving field state is transformed into a bound field state without the medium ever ceasing to be continuous.

The most important step for the particle observer is therefore to replace the word division. What looks like a division in the case of a wave is in truth a reorganization of the oscillation through interference. A wavefront is not a coherent lump that could be cut in half. It is a spatially distributed pattern of local field motions. Each point of the front is not a piece of the wave, but a local phase of a common state. When this pattern encounters a region in which the medium is differently tensioned and differently organized, then the pattern is not cut into two things, but converted as a whole into the allowed modes that the medium can support in that region. The medium does not take in the incoming motion as a package, but as a boundary condition. Where the oscillation encounters a strongly tensioned region, the original form of propagation can no longer remain unchanged. The result is not destruction, but mode conversion. Part of the motion can continue as a free oscillation, part couples into bound or strongly damped modes, part is deflected sideways, part is shifted into smaller scales. But all of this are not pieces of an object, but different outcomes of the same field-mechanical reorganization.

At this point a comparison helps that is not misleading. When a water wave runs into a reed field, it is not cut apart like a board. It is broken, scattered, decomposed into many smaller patterns, locally damped and at the same time newly re-emitted. Macroscopically one sees that the clear front disappears. Microscopically one sees that everywhere in the reeds new small waves arise. No one would say the water wave had been divided into two objects. One would say it has transitioned into a complex pattern of interference and dissipation. This is precisely the core of the UQSH transition. The black hole is not the point where waves stop, but the region where free propagation turns into bound organization. The wave does not end. It changes its form.

This also clarifies the second classical question, namely whether the free part of the wave simply passes the black hole. In the particle picture this is clear. A photon misses the hole or hits it. In the wave picture this is less clear, because a wave is spatially extended. The UQSH framework provides a clean answer. The wavefront is always a spherical response of the medium. It is distributed over large scales and has a local density. Where the medium curves strongly or is strongly prestressed, the local radiation density becomes field-mechanically compressed. The parts of the front that enter regions of higher coupling are more strongly reorganized and can transition into bound modes. Other parts of the front that pass through less coupled regions continue to propagate. But again, this is not a division of an object. It is a location-dependent transformation of a field state. That we describe this as division arises only because we treat the wavefront like a thing. In reality it is a state field that can react differently everywhere at the same time, depending on which field geometry it traverses.

Even if we wish to imagine a division, we can never observe it directly as observers of the same scale, because the nature of the reorganization itself is scale-nested. The UQSH view suggests that each interference is not simply a one-time superposition, but a cascade. Where an oscillation cannot simply be passed on, there is not an ending, but a new local response. This local response is again spherical, again isotropic on its own scale, and it in turn generates new small-scale deformations. A reception is itself again an emission. This means the wavefront does not hit a point and split there. It hits a region, and in that region countless smallest spherical reactions arise. The field responds to the arrival of the free boundary dynamics with a cloud of smaller boundary dynamics. Therefore macroscopically it appears like absorption, while microscopically it resembles a continued breaking up into ever finer organization.

Here also lies the key to the apparent paradox that a wave seems indivisible, yet one observes energy being absorbed locally. Within the UQSH framework indivisibility is not a property of an object, but a property of the level of description. On the level of free radiation the wave is a coherent pattern. As soon as it enters a strongly coupled regime, it does not lose this coherence abruptly, but gradually through interference, damping, and mode conversion. The pattern is not destroyed, but transformed into bound patterns. And bound patterns are precisely what we describe as matter or as the internal organization of a saturated state. Thinking this way also clarifies why interference never simply ends. There is no perfect boundary where a decision is suddenly made between continuing and stopping. There is only a transition region in which both exist simultaneously as a field response. Part of the motion remains in freely propagating modes, part becomes bound, and at the edges ever new smallscale spherical responses arise because the medium reacts locally wherever an adjustment is necessary.

For the particle observer it is helpful to condense this into one sentence. The black hole does not take a piece of the wave out as a chunk. It changes the conditions such that part of the incoming boundary dynamics is transformed into bound field organization. The rest continues to propagate, but not as the untouched original, rather as a newly formed interference pattern that emerges from contact with the saturated regime. One can imagine this like a river flowing into a swamp. The river is not divisible, but its course branches. One part seeps away. One part flows onward. And in the zone between them an entirely new pattern of vortices, channels, and backflows arises. In the UQSH sense this intermediate zone is what matters. That is where the physics takes place. Not in a hard event, but in the continuing cascade of interference and reorganization.

One may also adopt the formulation that at the boundary the wave is always between remaining and continuing. Not as a decision, but as a permanent field process. The boundary is not a geometric line, but a zone in which the medium can just still support free propagation, yet already begins to form bound modes. Therefore the wave is simultaneously continuing and bound, depending on which scale and which coupling one considers. And therefore in this picture there is no final division that an observer could see as a clear fact. There are only transitions, nesting, and the constant emergence of smaller spherical responses that weave themselves into the overall process.

4 Dissolution of Bound States in the Boundary Regime of Black Holes

An electron field node possesses no fixed internal substance and no point-like core. The dominant component of this state is a closed circulation at the boundary dynamics of the medium. This main circulation is accompanied by finer modes, by small-scale oscillations and phase structures that form on shorter scales and together ensure the stability of the overall organization. These are not separate vortices or components, but different resolution levels of the same motion. The electron is therefore not a composite object, but a nested resonance pattern of the field in which multiple scales are coherently coupled to one another.

This multiscale coupling simultaneously explains the inertia of bound states. Any change in motion requires the simultaneous reorganization of all participating field modes. The medium must not only be displaced locally, but readjusted across many scales. This collective reorganization produces a resistance to acceleration that appears macroscopically as mass. Inertia is thus not a property of a substance, but an expression of the bound field organization itself.

The stability of such a node requires that the medium can elastically absorb local deformations. The field must be sufficiently compliant to permanently sustain this closed circulation. Matter therefore exists only in tension regimes in which this elasticity is present. A particle is thus not an ontological ground state, but a dynamically stable special case of the field response. In the vicinity of a black hole, these conditions change continuously. With decreasing distance, curvature, tension density, and effective stiffness of the medium increase. The quantum foam approaches a saturated state in which additional local circulations can no longer be elastically supported. The conditions under which a closed vortex structure can stably exist are gradually restricted.

When bound field states such as electrons or positrons enter such a highly tensioned regime, they do not abruptly lose their existence, but gradually lose their coherence. First, outer portions of the circulation are modulated. Individual modes couple to the surrounding tension field and release energy. The closed motion increasingly opens. The stored energy progressively transitions into propagating relaxation modes. Eventually no stable binding remains, and the previously bound state transforms completely into free radiation.

Observationally, this transition appears as high-frequency emission in the X-ray and gamma spectrum. Matter is not destroyed in the process, but merely changes its form of organization. In field-mechanical terms, this is a regime change from standing to traveling waves, from bound to free dynamics. The black hole therefore does not act as a sink that swallows substance, but as a region in which bound states lose their stability and transition into relaxation motion.

It is essential that this process does not imply a sharp boundary or discontinuous division. An incoming radiation or matter wave is not split like an object. It interferes with the saturated tension state of the medium. Parts of the excitation couple into local relaxation modes, others continue to propagate. The dynamics remain continuous across all scales. What appears from a classical perspective as absorption or particle annihilation corresponds, field-mechanically, to an ongoing reorganization and redistribution of tension.

Free radiation is subject to different stability conditions than bound states. Since it possesses no closed structure from the outset, it can penetrate deeper into strongly tensioned regions before its propagation is modified. Bound states lose their coherence earlier. Matter therefore typically unbinds into radiation already in outer zones, before the innermost saturation regions are reached.

In this sense, black holes function as transformation zones of the field. They convert bound states into free excitations and couple stored energy back into large-scale dynamics. The boundary between matter and radiation thus does not appear there as a surface, but as a continuous transition zone in which binding passes into propagation.

5 Light Bubbles at the Double Slit and the Wave–Particle Dualism

The double-slit experiment appears puzzling only because the familiar conception of light unconsciously starts from a false picture. One imagines light as a collection of small particles, as tiny pellets that fly toward a screen like projectiles. Following this intuition, each of these pellets should be able to pass through only one of the two slits and subsequently possess a clearly defined trajectory. On the screen, two bright stripes should then appear, corresponding to the two openings.

Yet this is not what is observed. Instead, a characteristic interference pattern forms, consisting of many bright and dark stripes, as if a water wave had propagated through both openings simultaneously and then overlapped with itself. The result becomes even more astonishing when the intensity of the light is reduced to an extreme degree so that apparently only a single photon is traveling at a time. Even then, no two stripes appear on the screen. The individual impacts appear one after another as points, yet their distribution reconstructs over time exactly the same wave pattern.

This behavior seems paradoxical as long as one thinks in terms of particles. A single pellet cannot interfere with itself. Within the framework of the UQSH, however, this paradox disappears completely, because no particle is sent. Instead, a Light Bubble is generated. When a light source emits, no point-like object is launched. Rather, the medium begins to oscillate locally. This oscillation does not spread directionally but spherically in all directions at once. One may imagine it as a growing, pulsating bubble expanding outward from the emission site. This bubble is not located in space like an object; it is the motion of space itself.

When this spherical field motion reaches the double slit, something entirely natural occurs. The oscillation strikes the wall and is blocked there, except at the two openings. At those locations it can continue to propagate. The field motion therefore flows through both slits simultaneously. This does not happen because something splits like an object, but because the motion was spatially distributed from the very beginning. Every point of the wavefront continues its propagation. Behind the two openings, two new spherical field portions of the same oscillation arise and overlap in space.

Where the oscillations meet in the same phase, they reinforce each other. There the field excitation is stronger. Where they meet out of phase, they weaken each other. There the field excitation is weaker. In this way, a stable spatial pattern of zones with high and low activity forms, an interference carpet that determines the probability of interaction within the medium.

The screen does not measure the wave itself. It consists of atoms and thus of bound field nodes. These react only when the local field excitation is strong enough to trigger a transition. Only where the Light Bubble concentrates sufficient energy is an electron in the material excited or released. This event appears as a visible point. The point is therefore not the light itself, but the reaction of the material to a local field reorganization.

One may say that the wave is present over a large area, but only at certain locations does the medium switch into a new state. If light is now emitted many times in succession, each time a complete Light Bubble is formed. Each one again propagates through both slits and produces the same interference pattern in space. The screen, however, registers only a single local transition per emission, which is why only one point appears each time. Since each propagation generates the same geometric interference carpet, the points accumulate precisely where the field excitation is statistically strongest. After many repetitions, the invisible wave pattern becomes visible as a distribution of points.

The apparently random occurrence of individual impacts thus follows a clear underlying field geometry. There is no contradiction between wave and particle. The propagation is always wave-like and spatially distributed, while the measurement is always local. The wave belongs to the field, the point belongs to the detector. The so-called particle is nothing other than the moment in which an extended field motion triggers a stable reorganization at a small location.

Put simply, a Light Bubble strikes the wall and the medium locally switches into a bound state. It is not the light that becomes point-like, but the interaction that becomes point-like. In this way, the double-slit experiment loses its mystical character. It shows nothing supernatural, but merely that we are not dealing with small pellets, but with spatially extended field oscillations. The universe does not roll dice. It oscillates. Only the places where these oscillations leave traces in the medium become visible.

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